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Class: D20A

Roll no. 28

Aim: Create a Cryptocurrency using Python and perform mining in the Blockchain created.

Theory:

1. Blockchain Overview

Blockchain is a **distributed and decentralized ledger** that stores information in a series of linked blocks.

Each block contains:

- Transaction data
- Timestamp
- Previous block's hash
- Its own unique hash (digital fingerprint)

Once data is recorded in a blockchain, it becomes **immutable** because altering one block would require recalculating all subsequent blocks.

2. Mining

Mining is the process of:

1. Collecting pending transactions into a block.
2. Performing a computational puzzle (Proof-of-Work) to find a valid hash.
3. Adding the new block to the blockchain.
Broadcasting it to all connected peers.

Miners are rewarded with cryptocurrency for successfully mining a block.

3. Multi-Node Blockchain Network

In this lab, we simulate **three independent blockchain nodes** (5001, 5002, 5003).

Each node:

- Runs on a separate port.
- Maintains its own copy of the blockchain.
- Can connect with peers to share and validate blocks.

4. Consensus Mechanism

We use the **Longest Chain Rule**:

- If multiple versions of the chain exist, the **longest valid chain** is chosen.
- This ensures all nodes agree on a single transaction history.

5. Transactions & Mining Reward

Each transaction has:

- Sender
- Receiver
- Amount

When mining a block:

- Pending transactions are added to the block.
- A **reward transaction** is added automatically to pay the miner.

6. Chain Replacement

When `/replace_chain` is called:

1. Node requests chains from peers.
2. If it finds a longer and valid chain, it replaces its own.
3. This keeps the blockchain consistent across all nodes.

Tools & Libraries Used

- **Python 3.x**
- **Flask** – Web framework for API endpoints
`pip install Flask`
- **Requests** – For HTTP communication between nodes
`pip install requests==2.18.4`
- **Postman** – For testing API requests
- Python Standard Libraries:
 - `datetime`
 - `jsonify`
 - `hashlib`
 - `uuid4`
 - `urlparse`
 - `request`

Procedure and Output:

1. Download `hadcoin_node_5001.py`, `hadcoin_node_5002.py`, `hadcoin_node_5003.py`.
2. Install required packages.
3. Run each node in separate terminals.

```
Problems Output Terminal ... python + - [ ] [ ] ... | [ ] [ ]
(venv) PS C:\Users\Lenovo\Downloads\BC lab 3> pip install req
(venv) PS C:\Users\Lenovo\Downloads\BC lab 3> python hadcoin_
node_5001.py
* Serving Flask app 'hadcoin_node_5001'
* Debug mode: off
WARNING: This is a development server. Do not use it in a pro
duction deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5001
* Running on http://192.168.1.12:5001
Press CTRL+C to quit
[ ]
```

```
Problems Output Terminal ... python + - [ ] [ ] ... | [ ] [ ]
PS C:\Users\Lenovo\Downloads\BC lab 3> (Set-ExecutionPolicy -
Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\L
enovo\Downloads\BC lab 3\venv\Scripts\Activate.ps1")
(venv) PS C:\Users\Lenovo\Downloads\BC lab 3> python hadcoin_
node_5002.py
* Serving Flask app 'hadcoin_node_5002'
* Debug mode: off
WARNING: This is a development server. Do not use it in a pro
duction deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5002
* Running on http://192.168.1.12:5002
Press CTRL+C to quit
[ ]
```

```
Problems Output Terminal ... python + - [ ] [ ] ... | [ ] [ ]
PS C:\Users\Lenovo\Downloads\BC lab 3> (Set-ExecutionPolicy -
Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\L
enovo\Downloads\BC lab 3\venv\Scripts\Activate.ps1")
(venv) PS C:\Users\Lenovo\Downloads\BC lab 3> python hadcoin_
node_5003.py
* Serving Flask app 'hadcoin_node_5003'
* Debug mode: off
WARNING: This is a development server. Do not use it in a pro
duction deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5003
* Running on http://192.168.1.12:5003
Press CTRL+C to quit
[ ]
```

4. .Connect nodes using Postman – POST [/connect_node](#).

```
{
```

```
"nodes":  
  
[  
  
  "http://127.0.0.1:5002",  
  
  "http://127.0.0.1:5002"  
  
]  
  
}
```

The screenshot displays a REST client interface with the following components:

- URL Bar:** Shows the URL `http://127.0.0.1:5001/connect_node` with a `Save` button and a `Share` link.
- Method and URL:** A dropdown menu is set to `POST`, and the URL `http://127.0.0.1:5001/connect_node` is entered. A `Send` button is to the right.
- Tabs:** Below the URL bar are tabs for `Params`, `Auth`, `Headers (8)`, `Body` (selected), `Scripts`, `Tests`, and `Settings`. On the right, there are links for `Cookies` and `Beautify`.
- Body Tab:** The `Body` tab is active, showing a `raw` JSON payload:

```
{  
  "nodes": [  
    "http://127.0.0.1:5002",  
    "http://127.0.0.1:5003"  
  ]  
}
```
- Response Section:** Below the request body, the response is shown. It includes a status bar with `201 CREATED`, `5 ms`, and `325 B`. The response body is a JSON object:

```
{  
  "message": "All the nodes are now connected. The Hadcoin Blockchain now  
             contains the following nodes:",  
  "total_nodes": [  
    "127.0.0.1:5003",  
    "127.0.0.1:5002"  
  ]  
}
```

Workspaces More

SpecHub is your single home for API design and governance. Author specs visually, convert them to collections that stay in sync, and enforce standards across your team with custom governance rules. [Try SpecHub](#)

Overview **POST** http://127.0.0.1:5002/c/ + No environment

http://127.0.0.1:5002/connect_node Save Share

POST http://127.0.0.1:5002/connect_node **Send**

Params Auth Headers (8) **Body** Scripts Tests Settings Cookies Beautify

raw JSON

```
1 {
2   "nodes": [
3     "http://127.0.0.1:5001",
4     "http://127.0.0.1:5003"
5   ]
6 }
```

Body 201 CREATED • 5 ms • 325 B

{ } JSON Preview Visualize

```
1 {
2   "message": "All the nodes are now connected. The Hadcoin Blockchain now
3     contains the following nodes:",
4   "total_nodes": [
5     "127.0.0.1:5003",
6     "127.0.0.1:5001"
7   ]
8 }
```

Workspaces More

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Overview **POST** http://127.0.0.1:5003/c/ + No environment

http://127.0.0.1:5003/connect_node Save Share

POST http://127.0.0.1:5003/connect_node **Send**

Params Auth Headers (8) **Body** Scripts Tests Settings Cookies Beautify

raw JSON

```
1 {
2   "nodes": [
3     "http://127.0.0.1:5001",
4     "http://127.0.0.1:5002"
5   ]
6 }
```

Body 201 CREATED • 7 ms • 325 B

{ } JSON Preview Visualize

```
1 {
2   "message": "All the nodes are now connected. The Hadcoin Blockchain now
3     contains the following nodes:",
4   "total_nodes": [
5     "127.0.0.1:5002",
6     "127.0.0.1:5001"
7   ]
8 }
```

Console Postbot Runner Vault

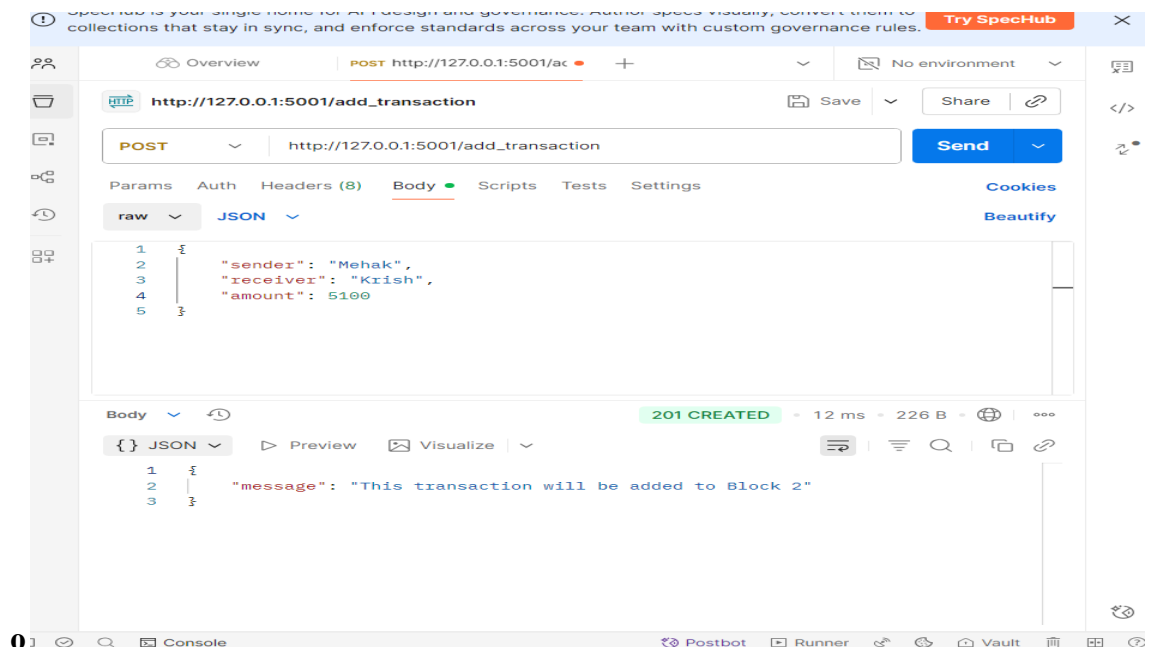
5. Add transactions – POST
6. Add 3 transactions in 5001
7. Condition 2 — Each node must have **different chain lengths**

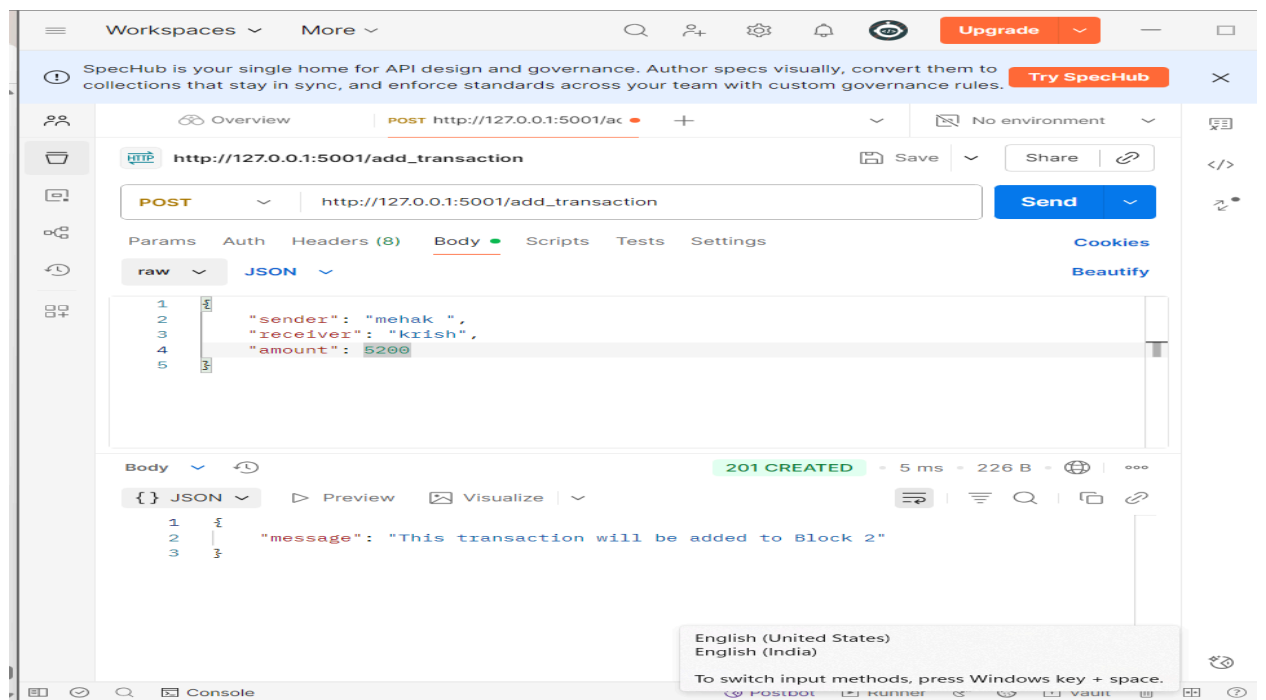
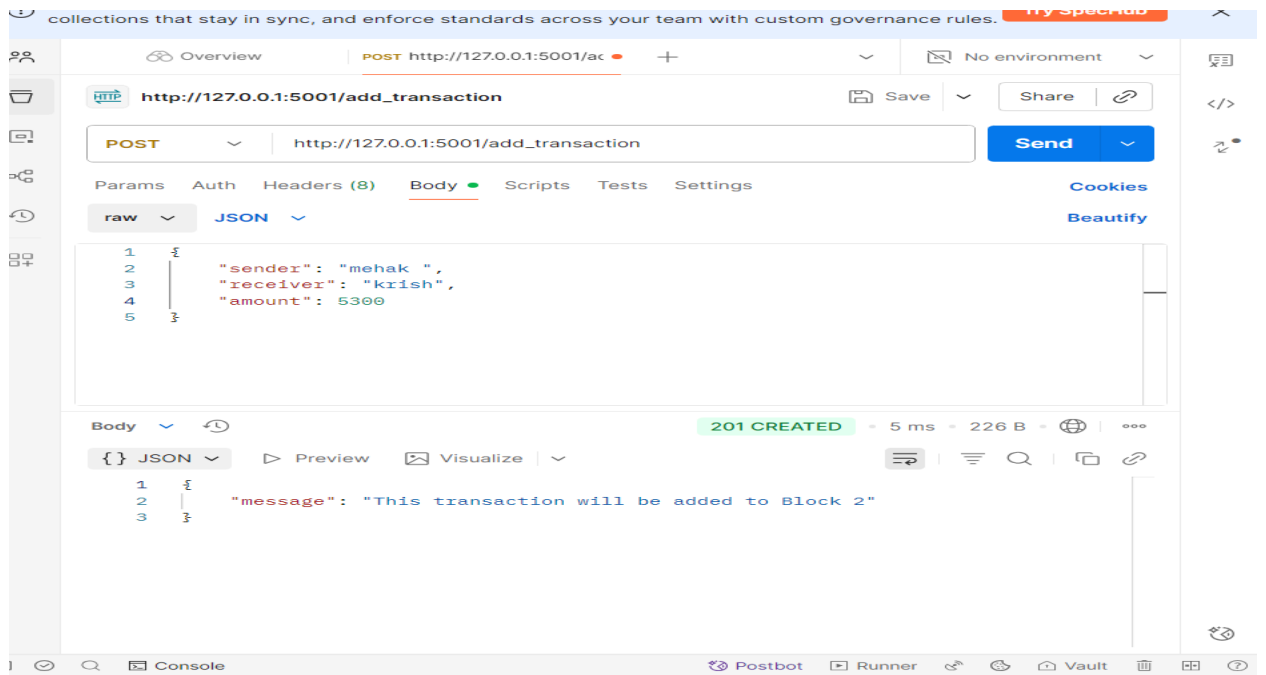
Example:

If you mined 1 block on 5001, but did not mine on 5002 or 5003 →

- 5001 chain length = 2
- 5002 chain length = 1
- 5003 chain length = 1

STEP 2 — Perform all transactions **ONLY** from one node (e.g., 5001)





GET

http://127.0.0.1:5001/mine_block

Send

Params

Auth

Headers (8)

Body

Scripts

Tests

Settings

Cookies

Body

200 OK • 8 ms • 620 B •

JSON

Preview

Visualize

```
1 {
2   "index": 2,
3   "message": "Congratulations, you just mined a block!",
4   "previous_hash":
5     "63bd533883994ec0805749ffc2569a63bf2e1628635336d45cf0cf1893b185d1",
6   "proof": 533,
7   "timestamp": "2026-08-30 20:42:49.443853",
8   "transactions": [
9     {
10      "amount": 5100,
11      "receiver": "Krish",
12      "sender": "Mehak"
13    },
14    {
15      "amount": 5200,
16      "receiver": "krish",
17      "sender": "mehak "
18    },
19    {
20      "amount": 5300,
21      "receiver": "krish",
22      "sender": "mehak "
23    }
24  ]
25 }
```

Console

Postbot

Runner

Vault

Params Auth Headers (8) Body • Scripts Tests Settings

Cookies

Body ▾ ↻

200 OK • 4 ms • 691 B • 🌐 • ⋮

{ } JSON ▾

▶ Preview

🖼 Visualize ▾



```
1  {
2    "chain": [
3      {
4        "index": 1,
5        "previous_hash": "0",
6        "proof": 1,
7        "timestamp": "2026-08-30 20:24:56.868502",
8        "transactions": []
9      },
10     {
11       "index": 2,
12       "previous_hash":
13         "63bd533883994ec0805749ffc2569a63bf2e1628635336d45cf0cf1893b18
14         5d1",
15       "proof": 533,
16       "timestamp": "2026-08-30 20:42:49.443853",
17       "transactions": [
18         {
19           "amount": 5100,
20           "receiver": "Krish",
21           "sender": "Mehak"
22         },
23         {
24           "amount": 5200
```



 http://127.0.0.1:5002/get_chain

Save

Share

GET

http://127.0.0.1:5002/get_chain

Send

Params

Auth

Headers (8)

Body

Scripts

Tests

Settings

Cookies

Body

200 OK

5 ms

290 B

{}

JSON

Preview

Visualize

```
1  {
2    "chain": [
3      {
4        "index": 1,
5        "previous_hash": "0",
6        "proof": 1,
7        "timestamp": "2026-08-30 20:25:39.617718",
8        "transactions": []
9      }
10   ],
11   "length": 1
12 }
```

collections that stay in sync, and enforce standards across your team with custom governance rules.

Overview

GET http://127.0.0.1:5003/ge

No environment

http://127.0.0.1:5003/get_chain

Save

Share

GET

http://127.0.0.1:5003/get_chain

Send

Params

Auth

Headers (8)

Body

Scripts

Tests

Settings

Cookies

Body

200 OK

5 ms

290 B

JSON

Preview

Visualize

```
1  {
2    "chain": [
3      {
4        "index": 1,
5        "previous_hash": "0",
6        "proof": 1,
7        "timestamp": "2026-08-30 20:25:52.848166",
8        "transactions": []
9      }
10   ],
11   "length": 1
12 }
```

Body

200 OK • 18 ms • 773 B •  • 

{ } JSON

Preview

Visualize

```
1 {
2   "message": "The nodes had different chains so the chain was replaced by
3   the longest one.",
4   "new_chain": [
5     {
6       "index": 1,
7       "previous_hash": "0",
8       "proof": 1,
9       "timestamp": "2026-08-30 20:24:56.868502",
10      "transactions": []
11    },
12    {
13      "index": 2,
14      "previous_hash":
15        "63bd533883994ec0805749ffc2569a63bf2e1628635336d45cf0cf1893b18
16        5d1",
17      "proof": 533,
18      "timestamp": "2026-08-30 20:42:49.443853",
19      "transactions": [
20        {
21          "amount": 5100,
22          "receiver": "Krish",
23          "sender": "Mehak"
24        }
25      ]
26    }
27  ]
28 }
```

 Console

 Postbot

 Runner



 Vault







ParamsAuthHeaders (8)Body • ScriptsTestsSettings

Cookies

Body

200 OK • 4 ms • 691 B •

{} JSONPreviewVisualize

```
11      "index": 2,
12      "previous_hash":
13        "63bd533883994ec0805749ffc2569a63bf2e1628635336d45cf0cf1893b18
14        5d1",
15      "proof": 533,
16      "timestamp": "2026-08-30 20:42:49.443853",
17      "transactions": [
18        {
19          "amount": 5100,
20          "receiver": "Krish",
21          "sender": "Mehak"
22        },
23        {
24          "amount": 5200,
25          "receiver": "krish",
26          "sender": "mehak "
27        },
28        {
29          "amount": 5300,
30          "receiver": "krish",
31          "sender": "mehak "
```

Workspaces ▾ More ▾

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Upgrade ▾

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Overview

GET http://127.0.0.1:5002

GET http://127.0.0.1:5002

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🔗 http://127.0.0.1:5002/get_chain

Save ▾

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GET ▾

http://127.0.0.1:5002/get_chain

Send ▾

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Params

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Headers (6)

Body

Scripts

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Body ▾

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200 OK

7 ms

691 B

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{ } JSON ▾

▶ Preview

🖼️ Visualize ▾

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1 {

2 "chain": [

3 {

4 "index": 1,

5 "previous_hash": "0",

6 "proof": 1,

7 "timestamp": "2026-08-30 20:24:56.868502",

8 "transactions": []

9 },

10 {

11 "index": 2,

12 "previous_hash":

13 "63bd533883994ec0805749ffc2569a63bf2e1628635336d45cf0cf1893b18

14 "5d1",

15 "proof": 533,

16 "timestamp": "2026-08-30 20:42:49.443853",

17 "transactions": [

18 {

19 "amount": 5100,

20 "receiver": "Krish",

21 "sender": "Mehak"

22 },

23 {

24 "amount": 5200,

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Console

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Workspaces ▾ More ▾

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👤 Overview GET http://127.0.0.1:5002/ GET http://127.0.0.1:5002/ + ▾ No environment ▾

🗑️ HTTP http://127.0.0.1:5002/get_chain Save ▾ Share 🔗

GET ▾ http://127.0.0.1:5002/get_chain Send ▾

Params Auth Headers (6) Body Scripts Tests Settings Cookies

Body ▾ ↺ 200 OK • 7 ms • 691 B • 🌐 ⋮

{ } JSON ▾ ▶ Preview 🖼 Visualize ▾

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```
17         "amount": 5100,
18         "receiver": "Krish",
19         "sender": "Mehak"
20     },
21     {
22         "amount": 5200,
23         "receiver": "krish",
24         "sender": "mehak "
25     },
26     {
27         "amount": 5300,
28         "receiver": "krish",
29         "sender": "mehak "
30     },
31     {
32         "amount": 1,
33         "receiver": "Richard",
34         "sender": "1d9edc0ff4b44a27bc0968ebeaac41d0"
35     }
36 ]
37 }
38 ],
39 "length": 2
40 }
```

📄 🔍 🗑️ Console 📧 Postbot ▶ Runner 🔔 🕒 🏠 Vault 🗑️ 🖼️ ?

Workspaces ▾ More ▾

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👤 Overview GET http://127.0.0.1:5003/replace_chain GET http://127.0.0.1:5003/replace_chain + ▾ No environment ▾

🗑️ HTTP http://127.0.0.1:5003/replace_chain Save ▾ Share

Send ▾

📄 GET http://127.0.0.1:5003/replace_chain

Params Auth Headers (6) Body Scripts Tests Settings Cookies

🕒 Body ▾ ↺ 200 OK • 13 ms • 773 B • 🌐 ⋮

{ } JSON ▾ ▶ Preview 🖼️ Visualize ▾

🔧 ⌵ 🔍 📄 🔗

17 {

18 "amount": 5100,

19 "receiver": "Krish",

20 "sender": "Mehak"

21 },

22 {

23 "amount": 5200,

24 "receiver": "krish",

25 "sender": "mehak "

26 },

27 {

28 "amount": 5300,

29 "receiver": "krish",

30 "sender": "mehak "

31 },

32 {

33 "amount": 1,

34 "receiver": "Richard",

35 "sender": "1d9edc0ff4b44a27bc0968ebeaac41d0"

36 }

37 }

38 }

39 }

40 }

🔍 Console Postbot Runner 🔗 🌐 Vault 🗑️ ?

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Overview

GET http://127.0.0.1:5001:5001

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HTTP

http://127.0.0.1:5001/is_valid

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http://127.0.0.1:5001/is_valid

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Body ▾

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{} JSON ▾

▶ Preview

🖼️ Visualize ▾

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1 {

2 | "message": "All good. The Blockchain is valid."

3 }

```

18
19 class Blockchain:
20
21     def __init__(self):
22         self.chain = []
23         self.transactions = []
24         self.create_block(proof = 1, previous_hash = '0')
25         self.nodes = set()
26
27     def create_block(self, proof, previous_hash):
28         block = {'index': len(self.chain) + 1,
29                 'timestamp': str(datetime.datetime.now()),
30                 'proof': proof,
31                 'previous_hash': previous_hash,
32                 'transactions': self.transactions}
33         self.transactions = []
34         self.chain.append(block)
35         return block
36
37     def get_previous_block(self):

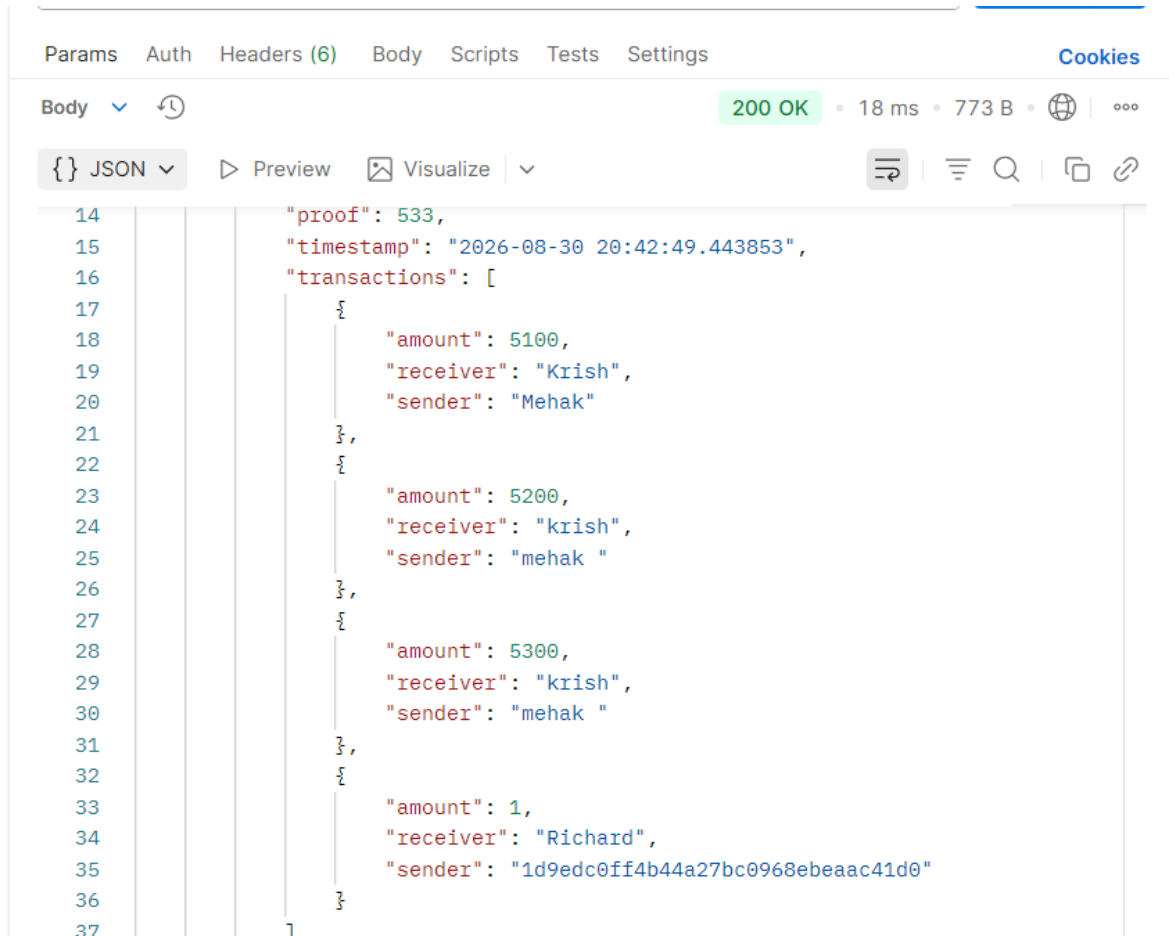
```

problems Output Terminal ... python + v [] [] ... [] [] []

```

(venv) PS C:\Users\Lenovo\Downloads\BC lab 3> python hadcoin_node_5003.py
127.0.0.1 - - [30/Aug/2026 20:38:04] "POST /connect_node HTTP/1.1" 201 -
127.0.0.1 - - [30/Aug/2026 20:45:46] "GET /get_chain HTTP/1.1" 200 -
127.0.0.1 - - [30/Aug/2026 20:48:41] "GET /get_chain HTTP/1.1" 200 -
127.0.0.1 - - [30/Aug/2026 20:50:34] "GET /get_chain HTTP/1.1" 200 -
127.0.0.1 - - [30/Aug/2026 20:51:01] "GET /replace_chain HTTP/1.1" 200 -

```



```
14      "proof": 533,  
15      "timestamp": "2026-08-30 20:42:49.443853",  
16      "transactions": [  
17          {  
18              "amount": 5100,  
19              "receiver": "Krish",  
20              "sender": "Mehak"  
21          },  
22          {  
23              "amount": 5200,  
24              "receiver": "krish",  
25              "sender": "mehak "  
26          },  
27          {  
28              "amount": 5300,  
29              "receiver": "krish",  
30              "sender": "mehak "  
31          },  
32          {  
33              "amount": 1,  
34              "receiver": "Richard",  
35              "sender": "1d9edc0ff4b44a27bc0968ebeaac41d0"  
36          }  
37      ]
```

Conclusion:

We developed a cryptocurrency using Python and implemented mining in a simulated three-node blockchain network. Each node maintained its own ledger and synchronized with peers using the Longest Chain Rule to ensure consistency. Mining was performed through Proof-of-Work, securely adding transactions and rewarding miners. This demonstrated key blockchain concepts, including decentralized consensus, transaction validation, and network synchronization.